CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Forth Semester of B. Tech. (CL) Examination

April 2013

CL 214 Fluid Mechanics-II

Date: 25.04.2013, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) What are the different dimensionless numbers which are important for fluid flow analysis? Define and explain any TWO of them. [04]

(b) Explain the following terms: Zone of silence, Compressibility correction factor, isothermal process. [03]

Q - 2 (a) Why do the basic equations for compressible fluid flow differ from those for the incompressible fluid flow? Obtain the energy equation for compressible flow for adiabatic process. [06]

(b) Explain: Mach angle, Mach number. Also obtain the relation between them. [08]

An aircraft flies at an altitude of 2000m where air temperature is 3^0 C. If its sound is heard 4.5 seconds after its passage over the head of a stationary observer on the ground, determine (i) Mach angle (ii) Mach number and (iii) distance covered by the aircraft between the instant it was directly overhead the observer and the instant he/she feels the disturbance due to the aircraft.

OR

Q - 2 (a) Classify the compressible flow based on Mach number. Discuss, with suitable sketch, the propagation of pressure wave when projectile velocity is less than the sound velocity. [06]

(b) What are the characteristics of the compressible flow at stagnation point? [08]

A jet airplane is propelled at a velocity of 1000 km/hr through air in which the atmospheric pressure is 88.3 kN/ m^2. The temperature is -5^0 C and the wind velocity is negligible. Calculate the pressure intensity, the temperature, the density of air at the stagnation point on the nose of the jet, and determine its Mach number. Take R = 290 Nm/kg ^{0}K and k = 1.4.

Q - 3 (a) Write short note on the following: **[06]**

(i) Distorted models (ii) Reynolds' model law

(b) The flow of liquid over a weir of included angle θ takes place due to gravity and the flow rate depend upon the head (**h**), density (**ρ**), and viscosity (**μ**). Derive the expression for the non-dimensional parameters that relates the flow rate (**Q**) to the independent variables. **[08]**

OR

Q - 3 (a) Using Buckingham's π-theorem, show that the resistance (F) to the motion of a sphere of diameter (D) moving with a uniform velocity (V) through a real fluid of density (ρ) and viscosity (μ) is given by: **[08]**

$$F = \rho D^2 V^2 . \Phi\left(\frac{\mu}{\rho V D}\right)$$

(b) What are the different similarities to be attained for the model to be dynamically similar to its prototype? Explain each of them in detail. **[06]**

SECTION – II

Q - 4 An oil of viscosity 9 poise and specific gravity 0.9 is flowing through a horizontal pipe of 60 mm diameter. If the pressure drop in 100 m length of the pipe is 1800 KN/m2, determine (1) the rate of flow of oil (2) the centre-line velocity (3) the total frictional drag over 100 m length (4) the power required to maintain the flow (5) the velocity gradient at the pipe wall (6) the velocity and shear stress at 8 mm from the wall. **[07]**

Q - 5 (a) Explain boundary layer separation and its control. Also discuss different methods for preventing the separation of boundary layer. **[07]**

(b) Derive equation for flow of viscous fluid between two plates when **[07]**

I. One plate is moving and other at rest

II. When both the plates are fixed.

OR

Q - 5 (a) Define boundary layer thickness and derive equations for displacement thickness, momentum thickness and energy thickness. **[07]**

(b) Derive Hagen –Poiseullie law with neat sketch. **[07]**

Q - 6 (a) A trapezoidal channel with side slopes of 1 to 1 has to be designed to convey 10 m^3/s at a velocity of 2 m/s so that the amount of concrete lining for the bed and sides is the minimum. Calculate the area of lining required for one metre length of canal. **[06]**

(b) Explain specific energy and specific energy curve. **[04]**

(c) Derive an expression for the discharge through a channel by Chezy's formula. **[04]**

OR

Q - 6 (a) Define hydraulic jump and derive equation for depth of hydraulic jump and loss of energy due to hydraulic jump. [08]

(b) Show that in a rectangular channel: [06]

1. Critical depth is two third of specific energy and
2. Froude number at critical depth is unity.

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